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6th OPJU International Technology Conference (OTCON 6.0) on SMARTICON (Smart & Intelligent Computing for Next-Generation Technologies) : Submission (343) has been created.

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Microsoft CMT <noreply@msr-cmt.org> Sun, Nov 1, 2026 at 10:30 PM
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The following submission has been created.

Track Name: Smart Computing for Data Science, Big Data & Analytics

Paper ID: 343

Paper Title: TRUF (Turf Booking and Sports Community Platform)

Abstract:
The rapid expansion of urban recreational sports facilities has highlighted significant operational inefficiencies in traditional booking workflows, including manual slot management, double-booking conflicts, fragmented player coordination, and financial losses due to last-minute team vacancies. Existing digital reservation platforms primarily focus on facility management while neglecting the social dynamics of participant coordination.
This paper introduces TRUF, a web-based integrated platform designed to bridge facility management and sports community engagement through an integrated substitute player recruitment mechanism. Built using a modern web architecture comprising Next.js, TypeScript, Tailwind CSS, PostgreSQL, and Prisma ORM, TRUF streamlines slot reservations, role-based access control, and automated participant management. The system introduces a state-driven recruitment module that identifies incomplete teams resulting from player withdrawals and opens vacant slots to compatible community members. Payment operations and notification dispatch workflows are integrated using Razorpay (Test Mode) with server-side HMAC-SHA256 signature verification and Resend transactional email services.
Deployed on the Vercel edge infrastructure, the application demonstrates enhanced slot utilization, transparent booking management, and streamlined substitute recruitment. The system architecture, database design, security mechanisms, and functional evaluations presented herein establish a scalable framework for digitalizing sports facility management while fostering sustainable recreational sports communities.
Keywords: Web Application Architecture, Sports Facility Reservation, Substitute Player Recruitment, Next.js, PostgreSQL, Database Transaction Control, Role-Based Access Control.

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Secondary Subject Areas: Not Entered

Submission Files:
TRUF International Research Publication.pdf (281 Kb, Sun, 01 Nov 2026 16:59:58 GMT)

Submission Questions Response: Not Entered

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